① $z = \frac{2+5i}{3+4i}$

a. Bentuk bilangan kompleks dalam bentuk $z = a + ib$

b. Nilai sekawan dari z

$$z = \frac{2+5i}{3+4i} \cdot \frac{3-4i}{3-4i} = \frac{(2+5i)(3-4i)}{(3+4i)(3-4i)}$$

$$= \frac{(2\cdot 3)+(2\cdot(-4i))+(5i\cdot 3)+(5i\cdot(-4i))}{3^2-(4i)^2}$$

$$= \frac{6-8i+15i-20i^2}{9-16i^2} = \frac{6+7i-20(-1)}{9-16(-1)}$$

$$= \frac{6+7i+20}{9+16} = \frac{26+7i}{25}$$

$$z = \frac{26}{25} + \frac{7}{25}i$$

Nilai sekawan : $z = \frac{26}{25} - \frac{7}{25}i$

② Bentuk kutub (polar) → $z = r(\cos\theta + i\sin\theta)$

r = modulus = $\sqrt{x^2+y^2}$

θ (Argumen) → $\theta = \arctan\left(\frac{y}{x}\right)$

2a. $z = (1+i)^7$ → Misal : $w = 1+i$

Bagian real $(x) = 1$

Bagian imajiner $(y) = 1$

Modulus : $r = \sqrt{1^2+1^2} = \sqrt{2}$

$\theta = \arctan\left(\frac{1}{1}\right) = 45^\circ$ (atau $\frac{\pi}{4}$ radian)

Bentuk kutub dari w adalah

$w = \sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$

Terapkan pangkat 7

Teorema De Moivre

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

$$z = w^7 = (\sqrt{2})^7(\cos(7\times 45^\circ) + i\sin(7\times 45^\circ))$$

$$\boxed{z = 8\sqrt{2}(\cos 315^\circ + i\sin 315^\circ)}$$

Nilai konjugat $(\bar{z})$ → Ubah tanda sudut

$$\bar{z} = 8\sqrt{2}(\cos(-315^\circ) + i\sin(-315^\circ))$$

KUADRAN

II	I
III	IV

Kuadran	x	y	θ°	Cara cari sudut
1	+	+	$0^\circ - 90^\circ$	$\arctan\left(\frac{\lvert y\rvert}{\lvert x\rvert}\right)$
2	−	+	$90^\circ - 180^\circ$	$180^\circ - \arctan\left(\frac{\lvert y\rvert}{\lvert x\rvert}\right)$
3	−	−	$180^\circ - 270^\circ$	$180^\circ + \arctan\left(\frac{\lvert y\rvert}{\lvert x\rvert}\right)$
4	+	−	$270^\circ - 360^\circ$	$360^\circ - \arctan\left(\frac{\lvert y\rvert}{\lvert x\rvert}\right)$

2b. $z = \frac{i\sqrt{2}}{3+3i}$ Ubah pembilang dan penyebut ke bentuk kutub

· Pembilang

Misal : $z_1 = i\sqrt{2}$

$z_1 = 0 + i\sqrt{2}$ (bentuk lengkap)

$x = 0$

$y = \sqrt{2}$ (positif)

$r_1 = \sqrt{0^2+(\sqrt{2})^2} = \sqrt{2}$

$\theta = 90^\circ$

$z_1 = \sqrt{2}(\cos 90^\circ + i \sin 90^\circ)$

tidak berada dalam kuadran manapun (tepat berada di sumbu-y positif)

· Penyebut

Misal : $z_2 = 3+3i$

$x = 3$, $y = 3$] kuadran I

$r_2 = \sqrt{3^2+3^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$

$\theta_2 = \arctan\left(\frac{3}{3}\right)$

$= \arctan(1)$

$= 45^\circ$

$z_2 = 3\sqrt{2}(\cos 45^\circ + i \sin 45^\circ)$

Operasi pembagian

$$z = \frac{z_1}{z_2} = \frac{\sqrt{2}}{3\sqrt{2}}(\cos(90-45) + i\sin(90-45))$$

$$z = \frac{1}{3}(\cos 45^\circ + i \sin 45^\circ)$$

$$\bar{z} = \frac{1}{3}(\cos(-45^\circ) + i \sin(-45^\circ))$$

$$\bar{z} = \frac{1}{3}(\cos 45^\circ - i \sin 45^\circ)$$

③ Diketahui

OP (titik pusat) $(0,0,0)$ → $A(2,4,2)$, $B(-1,1,3)$, $C(0,3,2)$

$\overline{AB}$ wakil vektor u

$\overline{BC}$ wakil vektor v

Vektor satuan A $(\hat{a})$

$$|A| = \sqrt{2^2+4^2+2^2} = \sqrt{4+16+4} = \sqrt{24} = 2\sqrt{6}$$

$$\hat{a} = \frac{1}{2\sqrt{6}}(2,4,2) = \left(\frac{2}{2\sqrt{6}}, \frac{4}{2\sqrt{6}}, \frac{2}{2\sqrt{6}}\right)$$

$$= \left(\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$= \left(\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{6}\right)$$

Vektor Satuan B

$|B| = \sqrt{(-1)^2+1^2+3^2} = \sqrt{1+1+9} = \sqrt{11}$

$\hat{b} = \frac{1}{\sqrt{11}}(-1,1,3) = \left(-\frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{3}{\sqrt{11}}\right)$

$\hat{b} = \left(-\frac{\sqrt{11}}{11}, \frac{\sqrt{11}}{11}, \frac{3\sqrt{11}}{11}\right)$

Vektor Satuan C

$|c| = \sqrt{0^2+3^2+2^2} = \sqrt{0+9+4} = \sqrt{13}$

$\hat{c} = \frac{1}{\sqrt{13}}(0,3,2) = \left(0, \frac{3}{\sqrt{13}}, \frac{2}{\sqrt{13}}\right)$

$\hat{c} = \left(0, \frac{3\sqrt{13}}{13}, \frac{2\sqrt{13}}{13}\right)$

Panjang u dan v

Panjang vektor u (wakil AB) $AB = B - A$

$u = AB = (-1-2, 1-4, 3-2)$

$u = (-3,-3,1)$

$|u| = \sqrt{(-3)^2+(-3)^2+1^2}$

$= \sqrt{9+9+1}$

$|u| = \sqrt{19}$

Panjang vektor v (wakil BC) $BC = C - B$

$v = BC = (0-(-1), 3-1, 2-3)$

$v = (1,2,-1)$

$|v| = \sqrt{1+4+1}$

$|v| = \sqrt{6}$

4. $R = 2i + j - 3\hat{k}$

$S = 2i + 5j + ak$

$T = 2i + j - \hat{k}$

Besar vektor R (|R|)

$|R| = \sqrt{2^2+1^2+(-3)^2}$

$= \sqrt{4+1+9}$

$|R| = \sqrt{14}$

Panjang vektor A = RS

$A = S - R$

$= (2i + 5j + ak) - (2i + j - 3k)$

$= (2-2)i + (5-1)j + (a-(-3))\hat{k}$

$= 0\hat{i} + 4j + (a+3)\hat{k}$

$|A| = \sqrt{0^2+4^2+(a+3)^2}$

$= \sqrt{0+16+(a^2+6a+9)}$

$= \sqrt{16+(a^2+6a+9)}$

$|A| = \sqrt{a^2+6a+25}$

Perkalian vektor $B = R \cdot (S \times T)$

$B = R \cdot (S \times T)$

$= R \cdot (2i + 5j + ak) \times (2i + j - k)$

$= R \cdot (2i \times 2i) + (2i \times j) + (2i \times (-k)) + (5j \times 2i) + (5j \times j) + (5j \times (-k)) + (ak \times 2i) + (ak \times j) + (ak \times (-k))$

$B = R \cdot 0 + 2k + 2j - 10k + 0 - 5i + 2aj - ai$

$B = R(-5-a)\hat{i} + (2+2a)j - 8k$

$B = (2)(-5-a) + (1)(2+2a) + (-3)(-8)$

$B = -10 - 2a + 2a + 24$

$B = -10 + 2 + 24$

$B = 16$

Sudut yang dibentuk oleh vektor R dan S

$\cos\theta = \frac{R \cdot S}{|R||S|}$

$= \frac{(2)(2)+(1)(5)+(-3)(0)}{\sqrt{14} \cdot \sqrt{2^2+5^2+a^2}}$

$= \frac{4+5-3a}{\sqrt{14} \cdot \sqrt{4+25+a^2}}$

$= \frac{9-3a}{\sqrt{14} \cdot \sqrt{29+a^2}}$

$\cos\theta = \frac{9-3a}{\sqrt{14}\sqrt{29+a^2}}$

$\theta = \arccos\left(\frac{9-3a}{\sqrt{14(29+a^2)}}\right)$